

Round Two Deliverables

Submission report and evidence pack

Project	PayChain — M-PESA merchant payments platform (Kenya)
Report date	21 September 2026
Prepared for	Stellar Community Fund reviewers and evaluators
Network	Stellar testnet (horizon-testnet.stellar.org). No mainnet funds are involved.
Repository	github.com/PayChainKE/paychainkeHQ , branch staging
Commits	949faee backend, ledger and stress test 042243c merchant UI 473dceb pilot script and evidence document
Evidence run	20 September 2026, 20:52 to 20:55 UTC — Stellar ledgers 4,782,793 to 4,782,829

36 transactions confirmed on Stellar testnet	36 / 36 independently re-verified on Horizon	50 operations across 5 pilot merchants	511.58 USDC auto-settled (22 settlements)	310.26 USDC redeemed (14 payouts)
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Every amount and hash in this report was produced by the running PayChain backend against the public Stellar testnet and then checked against Stellar Horizon. The fiat leg of the redemption flow (the M-PESA B2C disbursement) is a **simulation**, as specified in the deliverable; nothing in this evidence moves real money.

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1. Executive summary

Round two adds two capabilities on top of the round one Stellar wallet: merchants can decide how much of each incoming M-PESA payment is routed to their Stellar wallet (Deliverable 1), and they can redeem Stellar funds back to local fiat on demand (Deliverable 2). Both are implemented, tested end to end, and evidenced on the public testnet. A 50-operation stress test across five pilot merchants produced 36 confirmed testnet transactions, each of which was independently re-checked on Horizon.

The two deliverable items that still need to be attached to complete the submission are the demo video and the pull-request links (Section 8). They are listed openly rather than implied as done.

Item	Status	Evidence
Deliverable 1: settlement rule configuration	Implemented and verified	Sections 3 and 5; Appendix B
Deliverable 2: withdraw / redeem flow	Implemented and verified	Sections 4 and 5; Appendix B
Redemption UI and settlement statement export	Implemented; screen capture pending	Section 4; CSV export verified by test
Pilot stress test: 50 operations, 5 merchants	Complete	Section 5.3; Appendix B
Error handling: low balance, network timeout	Verified	Section 6
20+ verified testnet transaction hashes	Complete (36)	Appendix B
End-to-end video walkthrough	Pending	To be attached
GitHub pull-request links	Pending	Commits are on branch staging; PR links to be attached

2. Scope and approach

- **Testnet only.** All on-chain activity is on the Stellar testnet using the project USDC test asset. Nothing touches mainnet.
- **Simulated fiat leg.** The M-PESA B2C disbursement is a locally generated, Daraja-shaped request and result. It is stored for audit but is never sent anywhere; its callback URLs use the reserved `.invalid` domain, so they cannot resolve.
- **Pilot merchants only.** The features are enabled only for accounts an administrator has flagged as Stellar pilot accounts. Regular merchants receive a "not available" response from every new endpoint and are unaffected on the payment path.
- **Delivered on the PayChain demo dashboard.** The merchant screens live in the demo application (Inflation Shield page). The production merchant dashboard and mobile app were deliberately left unchanged.

3. Deliverable 1: settlement rule configuration

"Backend settlement rule configuration that enables merchants to define how incoming M-PESA payments are routed, with a UI settings toggle on the merchant dashboard."

Requirement	Implementation	Where
Percentage-split fields on the merchant schema	A per-merchant rule: on/off and the percentage routed to Stellar (0 to 100). The liquid share is the remainder.	<code>models/Merchant.js</code> (<code>settlementRule</code>)
Backend calculates split ratios on incoming payloads	After a payment is credited in KES, the configured share is converted to testnet USDC and sent to the merchant wallet. Amounts are rounded to cents; the two parts always add up to the credit.	<code>services/stellarSettlementService.js</code> ; hooked into the NCBA credit path and both STK settlement paths

Requirement	Implementation	Where
Slider on the merchant settings UI	On/off switch, 0 to 100% slider in 5% steps, a live preview ("if a customer pays Ksh 10,000: X stays liquid, Y goes to Stellar") and a Save action with confirmation.	apps/demo/.../SettlementRuleCard.jsx
Save custom allocation percentages	Saved through PUT /api/transactions/settlement-rule; values outside 0 to 100, non-numeric input and injection attempts are rejected.	Verified by API tests (Section 5.2)

Design decisions

- **Never on the payment path.** The conversion runs after the KES credit and detached from it, so a slow Stellar network cannot delay or fail a payment or its notification.
- **Exactly once.** A unique ledger key allows only one settlement per incoming payment, so a retried webhook cannot split the same payment twice.
- **Conservative debit.** The KES debit is an atomic conditional update of exactly the amount just credited; it cannot overdraw the balance.

4. Deliverable 2: withdraw / redeem

"A one-click Withdraw / Redeem flow that debits the merchant's Stellar testnet wallet and triggers an automated M-PESA B2C simulated disbursement to the merchant's registered phone number."

Requirement	Implementation	Where
Verify balance, sign a debit, generate a receipt hash	Reads the live on-chain USDC balance, rejects a request above it, then signs and submits a Stellar payment from the merchant wallet. A SHA-256 receipt hash is computed over the canonical (key-sorted) receipt fields.	redeemToMpesa in the settlement service
Simulated Safaricom B2C payload and response	A Daraja-shaped BusinessPayment request and a successful result (ResultCode 0, marked SIMULATED) are stored on the ledger entry, addressed to the merchant's registered phone (shown masked).	buildSimulatedB2C
"Request Payout" modal with loading and success states	Amount entry with Max, live KES preview, progress state while the wallet is debited, and a success receipt with the StellarExpert link, M-PESA receipt and receipt hash.	RequestPayoutModal.jsx
Downloadable reconciliation table linked to StellarExpert	Every settlement and payout appears with type, status, KES, USDC, transaction link and receipt hash, plus a CSV download whose rows carry the StellarExpert URL.	ReconciliationTable.jsx; GET /wallet/reconciliation(.csv)

Safeguards

- A double-click or immediate identical retry of a payout is refused for 20 seconds.
- The CSV export neutralises spreadsheet formula injection (cells beginning with = + - @ are escaped).
- The payout endpoint is rate limited and available to pilot accounts only.

5. Verification and test results

5.1 Automated tests

The backend suite passes in full: **61 of 61** tests (51 existing plus 10 added for this work). The new tests cover split arithmetic and rounding, receipt-hash determinism, the simulated B2C payload (including that it cannot be sent

anywhere), CSV escaping and StellarExpert links, and the classification of definitive versus ambiguous Stellar failures.

5.2 API acceptance checks

Twenty checks were run against the real routes with real login tokens and a controlled chain: **all 20 passed**.

- An unauthenticated caller is refused; a regular (non-pilot) merchant is refused on all five new endpoints.
- The rule rejects 150%, -1%, a non-boolean switch and an operator-injection value; a valid rule saves and persists.
- A payout above the wallet balance, of zero, or of garbage input is rejected; a valid payout returns a complete receipt (hash, StellarExpert URL, simulated B2C, masked phone).
- An identical immediate retry is blocked; the statement lists the payout and the CSV downloads with the correct headers.

5.3 Pilot stress test: 50 operations across 5 merchants

The script backend/scripts/stellar-stress-test.js drives the real credit path and the real redemption service. It was run once against an in-memory chain and once against the live Stellar testnet; both runs passed every check. The figures below are from the live run. Five pilot accounts were used, each with its own wallet and rule; the fifth is a control with the rule switched off.

Pilot	Rule	Settlements	Payouts	Final wallet (USDC)	Notes
1	20% to Stellar	6	4	26.5489	
2	50% to Stellar	5	3	41.2950	One payment hit an injected "master wallet underfunded" fault
3	80% to Stellar	5	4	103.9760	One payment hit an injected network timeout
4	20% to Stellar	6	3	29.4988	One payout hit an injected network timeout
5	Off (control)	0	0	0.0000	Received 6 payments; none settled, as required

Operation group	Count	Outcome
Incoming payments (30)	22	Settled to Stellar; on-chain transaction recorded
	6	Control merchant: rule off, kept fully liquid
	1	Injected network timeout: recorded as unconfirmed , not refunded (see Section 6)
	1	Injected underfunded master wallet: recorded as failed , KES returned
Payout requests (20)	14	Completed: on-chain debit, receipt hash, simulated B2C
	3	Control merchant has no USDC: rejected with INSUFFICIENT_USDC
	2	Request above balance: rejected with INSUFFICIENT_USDC
	1	Injected network timeout: reported to the caller, not treated as success

After the run, the script re-checks these invariants for every merchant and all held: **(1)** KES is conserved (credits minus amounts sent to Stellar); **(2)** the on-chain USDC balance of every wallet equals the ledger (settled minus redeemed); **(3)** every completed entry has a transaction hash and a receipt hash, and the hashes recompute; **(4)** the control merchant has no entries; **(5)** a replayed webhook is rejected and a replayed split is a no-op; **(6)** the CSV contains StellarExpert links.

5.4 Independent on-chain verification

All 36 transaction hashes were re-fetched from Stellar Horizon after the run. **36 of 36** exist, were successful, and are single USDC payment operations; each amount on chain equals the amount in our ledger to seven decimals. They span ledgers 4,782,793 to 4,782,829 (20 September 2026, 20:52:32 to 20:55:32 UTC). The complete table is in Appendix B.

6. Error handling: low balance and network timeouts

The deliverable asks for verified handling of low balances and RPC timeouts. The behaviour below was exercised in the stress test.

Condition	Behaviour	Reason
Payout above the wallet balance	Rejected before anything is signed (INSUFFICIENT_USDC). No state changes.	The on-chain balance is read first.
Stellar network does not respond in time	Recorded as unconfirmed with a clear message. KES is not automatically returned; a payout reports an error and asks the merchant to check the balance before retrying.	The transaction may still land. Refunding automatically could pay the merchant twice.
Definitive rejection (for example master wallet underfunded, missing trustline)	Recorded as failed and the merchant's KES is returned in full.	Nothing reached the ledger, so a refund is safe.
Duplicate payment webhook	Rejected; a replayed split is a no-op.	One settlement per incoming payment.
Repeated identical payout request	Refused for 20 seconds.	Prevents accidental double payouts.

7. How a reviewer can verify this independently

- **Any transaction:** open its StellarExpert link in Appendix B, or query <https://horizon-testnet.stellar.org/transactions/<hash>>. It shows a successful USDC payment between the master wallet and the pilot wallet.
- **Any pilot wallet:** open its account page (Appendix A) to see its USDC balance and full payment history.
- **Re-run the stress test:** `node backend/scripts/stellar-stress-test.js --mode=live` against a local database. The script refuses to run against any non-local database and prints its full report with links.
- **Code:** the requirement-to-file mapping is in `docs/stellar-instaward-round2.md` in the repository.

8. Disclosures and outstanding items

Stated plainly so the evidence is not read as claiming more than it shows.

Item	Detail
Testnet and simulation	All on-chain evidence is testnet. The M-PESA disbursement is simulated by design and never sent.
Fixed exchange rate in the stress test	To make the statements reproducible, the stress test used a fixed rate of 130 KES per USD. The product itself uses the live rate.
Test accounts	The five stress-test pilots were generated in an isolated local database. The named, realistic pilot merchants are created with the provided script in the live environment.
Screen verification	The interface components compile and their API calls are tested, but an in-browser capture of the slider, modal and table has not yet been made. It will be included in the video.
STK settlement hooks	The split hook on the NCBA credit path is exercised end to end by the stress test. The two STK settlement hooks share the same function and passed syntax and import checks, but were not run end to end.
Demo video	Pending. To be recorded and attached.

Item	Detail
Pull-request links	Pending. The work is committed on branch staging (commits listed on the cover); pull-request links to be attached. Repository access may need to be granted to reviewers.

Appendix A. Pilot wallets (Stellar testnet)

Public keys only. Each link opens the account on StellarExpert.

Pilot	Rule	Account (public key)
1	20% to Stellar	GBQ2TDAW3BHX6U27ULBYXB2I3GMJNFE0A4BNZRDC4JM7XQFA3XG7FSXR
2	50% to Stellar	GCGZFQIJDB5WTPMS0AWZAR3YIG6L26EDUVHIX0QSHASZ45RZP5K0YBSC
3	80% to Stellar	GAHB2JWQS5R5EHVK6WIEGRRF30V4ZTZJG3J6A7TIB5VLDRT6GDRFKE5N
4	20% to Stellar	GBWFHLDSES7ZDX0JALN3PPYKPMMT6V42SEK2Q6V7KVST6Y5GJJLTPYI6
5	Off (control)	GA5FPB2N3P2JY470J53GMFSM4XGH6VHONXXS2Z47GMB4SDFILU0SFABB

Appendix B. Confirmed transactions (36)

Type: **Settlement** = share of an incoming payment converted to USDC and sent to the pilot wallet; **Payout** = USDC debited from the pilot wallet for redemption. All transactions were confirmed on 20 September 2026; times are UTC. Each hash links to StellarExpert (testnet).

#	Pilot	Type	USDC	Ledger	Time UTC	Transaction hash
1	1	Settlement	2.3076923	4,782,793	20:52:32	11d70afef55258cc450724d46cee9e1d5665903396a322fd8cedf1dc034c5d4c
2	2	Settlement	12.3076923	4,782,794	20:52:37	96e98e6e5314bc94456b28f147b3db19ee511ba56949c49a363fda47efbe0926
3	3	Settlement	48	4,782,795	20:52:42	b42a1572b22b043b78a99993af82eb605e8d28a42ba8429e709efdaf9531a7f8
4	4	Settlement	18.4615385	4,782,796	20:52:47	c3c5231728a5f2ba3f3d813b4942d17cd14245a7f922a436b5499f811865baa7
5	1	Settlement	4.9230769	4,782,797	20:52:52	a206f115e7a9f448517209f45f8a52f079115f5359a98fe51d058ddeee2f6174
6	2	Settlement	30	4,782,798	20:52:57	235527ea030dc3e02dd10579b9c191b202e8e423cd3eb1209f0ebbf8f876bc73
7	3	Settlement	73.8461538	4,782,799	20:53:02	bc931fbbcffdaaf68cdf1bd0a4196506d781491df0f15741f16ad8c398007f81
8	4	Settlement	1.4615385	4,782,800	20:53:07	1202750ce45f499111ffc24979f278f79b0d527c609de87f2ec02ba6170775fd
9	1	Settlement	12	4,782,801	20:53:12	8b3724ad23e2039dd1d3a655e3bc435a9e9a5460dcc612eb973d4a10d4610467
10	2	Settlement	46.1538462	4,782,802	20:53:17	b3c54fc1ff29378f16c0c4d3ee62641b6d93cf17b6859100e937b4c9a305cd4f
11	4	Settlement	30.7692308	4,782,803	20:53:22	ff2a33a51887f23a6486fa81ae21c6a2f05b62652899cfe0222dac402a79e09e
12	1	Settlement	18.4615385	4,782,804	20:53:27	3ea3deef3bd9b7d9c7444f7b79c91069947131c7477fb64deefad64c3c21997e
13	2	Settlement	3.6538462	4,782,805	20:53:32	3d745e690ad26e290962e8d6c08486bb9734bfcc8473ec856e7a00096bed9476
14	3	Settlement	123.0769231	4,782,806	20:53:37	89984a460f78bed337140f2de3a830df645792ca4c3d32de03d2a71b582d9667
15	4	Settlement	2.3076923	4,782,807	20:53:42	8de2bf0d4e76e4ea2c9abd96ac3dc8ae05c9acc79c50682be18e77186be38da8
16	1	Settlement	1.4615385	4,782,808	20:53:47	94005b6918b823bbe14c271c76a68eeb36ca113fd185b245a491f0952ddcab74
17	3	Settlement	9.2307692	4,782,809	20:53:52	c3bbb42b5d208f4c53182c357e92c08f54e22283132904804fc36b811ceafa9b
18	4	Settlement	4.9230769	4,782,810	20:53:57	2b8f85e88b32952ae2ee67e808de9426000e49bbce8c259d1c846a19b15adc35

#	Pilot	Type	USDC	Ledger	Time UTC	Transaction hash
19	1	Settlement	30.7692308	4,782,811	20:54:02	63f7a6e677e998676582bca179b093c109a3cbc9f0a479011ef796256a9abeb5
20	2	Settlement	5.7692308	4,782,812	20:54:07	af05a92f8a89f929f664bd8b47c751548484a9ccc19c0806f8488315e960b8b1
21	3	Settlement	19.6923077	4,782,813	20:54:12	81fc8a80f4332ee8b7808d2d53a8ff2442b9ef9cb62e12cc43f180ced71111a7
22	4	Settlement	12	4,782,814	20:54:17	1e130c7adf99d849dd324737cfff78f27fec89c8c6d0f0f9f4dfd471e3d2a137
23	1	Payout	17.4808	4,782,815	20:54:22	7e8d9fdac15be82404e619f3857e8149ccabd47774f525ec3207266056e2e1e5
24	1	Payout	13.1106	4,782,816	20:54:27	ed0681089eb070bdc57c408078bbe2d8e67fd77cb2d99572959bed4631b06bc
25	1	Payout	9.8329	4,782,817	20:54:32	881f361abb37629b8502a0d869560a2850fda314b9ca32224ef0ec3ca17f2f66
26	2	Payout	24.4712	4,782,818	20:54:37	51580f83c7bcd528e91949c689c1a847005b112477eab8117486c149a5890c0
27	2	Payout	18.3534	4,782,819	20:54:42	6a9acf7d78cbb5e921102f740bc2b434fcfbaf6d1369ff07b8cc2cdf20eeb22f
28	2	Payout	13.765	4,782,820	20:54:47	6635aa37672b1b1da5ff6672460dd4adab35c0f32c004824d3a854ea68c35c86
29	3	Payout	68.4615	4,782,821	20:54:52	233ada0c78867554661d5f0863b3669f8c3bfc5973f04f772fd3e478d5a82d2e
30	3	Payout	51.3462	4,782,822	20:54:57	fc0d75ba08facd1fc393def0953fb2cfa2eadfd87e5a313f8275f3b4da1a687a
31	3	Payout	38.5096	4,782,823	20:55:02	5af7feac19d308af683b74e6eb1d958779cbf6ad8014f095a0fbcd9d114645c3
32	4	Payout	17.4808	4,782,824	20:55:07	9dc0202f635b9199d29f697e376f7b74808b2fb6f92dc7baafbba77895a08bdd
33	4	Payout	13.1106	4,782,825	20:55:12	0e4c5a074eb71bcf10f3f52cae313c064f5e6e223476b9a5d54092f00b750f41
34	4	Payout	9.8329	4,782,826	20:55:17	395ab4c274745131492fd66b944dcccffbe121a1e67c3bc2e671091f497315623
35	1	Payout	2.9499	4,782,828	20:55:27	d432cd7134c0b3dd63c7262a07d697128404d58003b3c05468a370a42c77b4ce
36	3	Payout	11.5529	4,782,829	20:55:32	acfbcb7673e52156c4b21011025815a0ab1aeb95097ecea67b70a4a7a5758ba5f